



Is Bone Morphogenetic Protein The New Gold Standard?

A Meta-Analysis and Systematic Review of Bone Morphogenetic Protein vs Autologous Iliac Crest Bone Graft Use in Spinal Fusion

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INTRODUCTION

Bone Grafting



Promote successful
spinal fusion

Autogenous iliac Crest Bone Graft (ICBG)

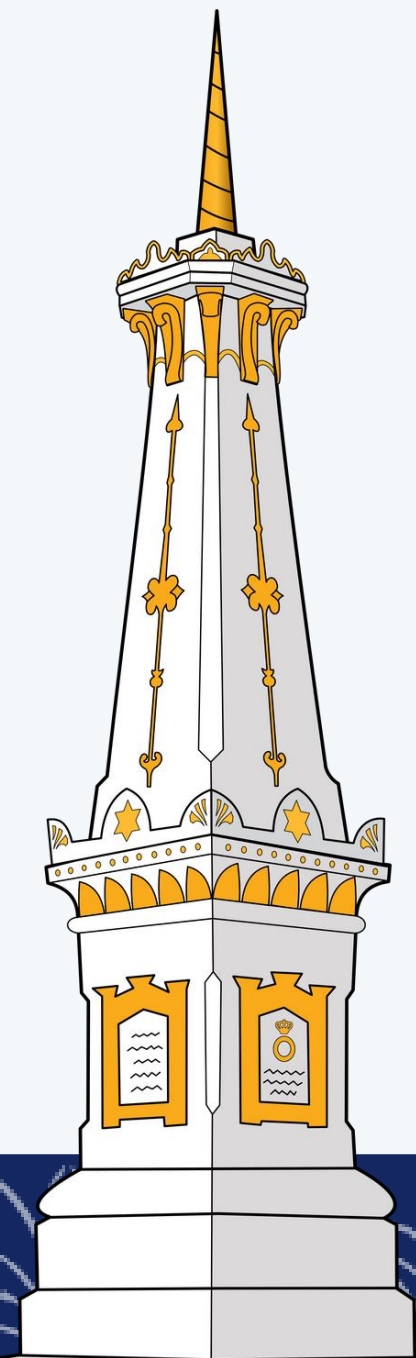
Standard choice for cervical and lumbar
fusion procedure
(-) Donor site pain, quality of graft depends
on age and health of patient

Allograft

(-) less effective osteoinductive function,
carries risk of disease transmission, such
as Hepatitis B, C, and HIV

Synthetic Bone Graft

Bone Morphogenetic Proteins (BMP)

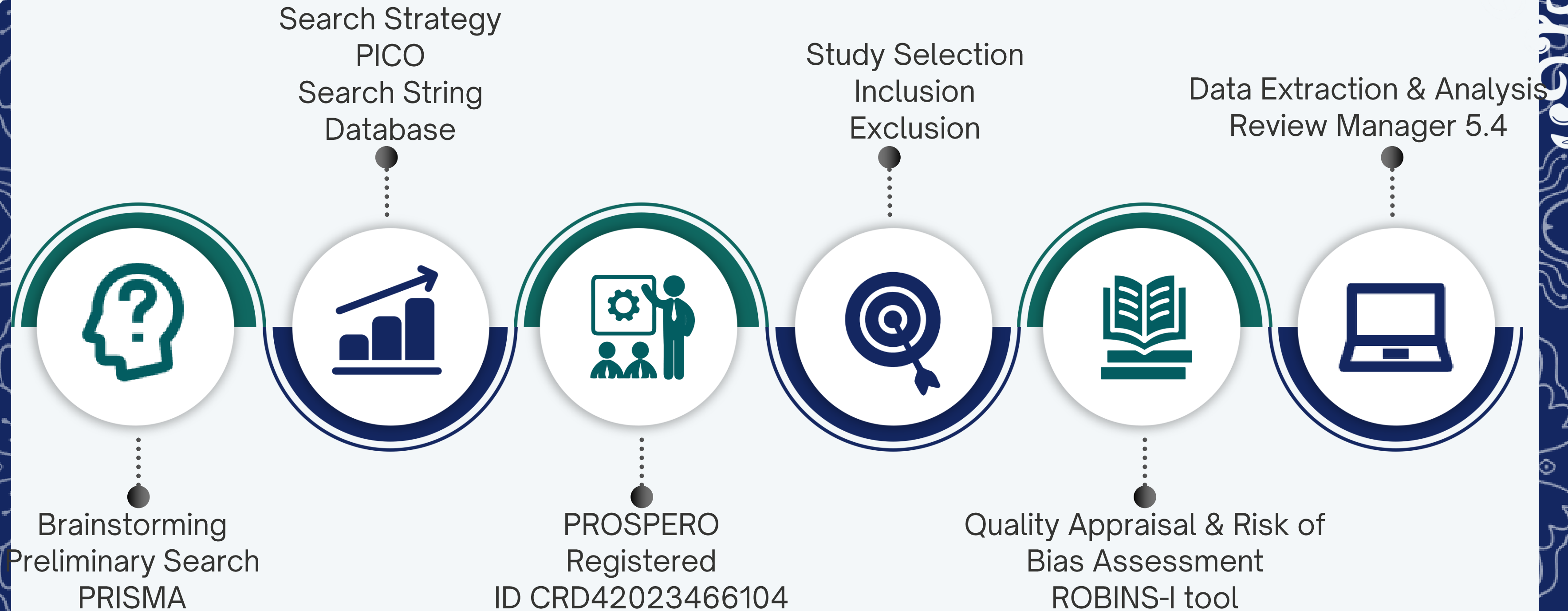


BONE MORPHOGENETIC PROTEIN (BMP)

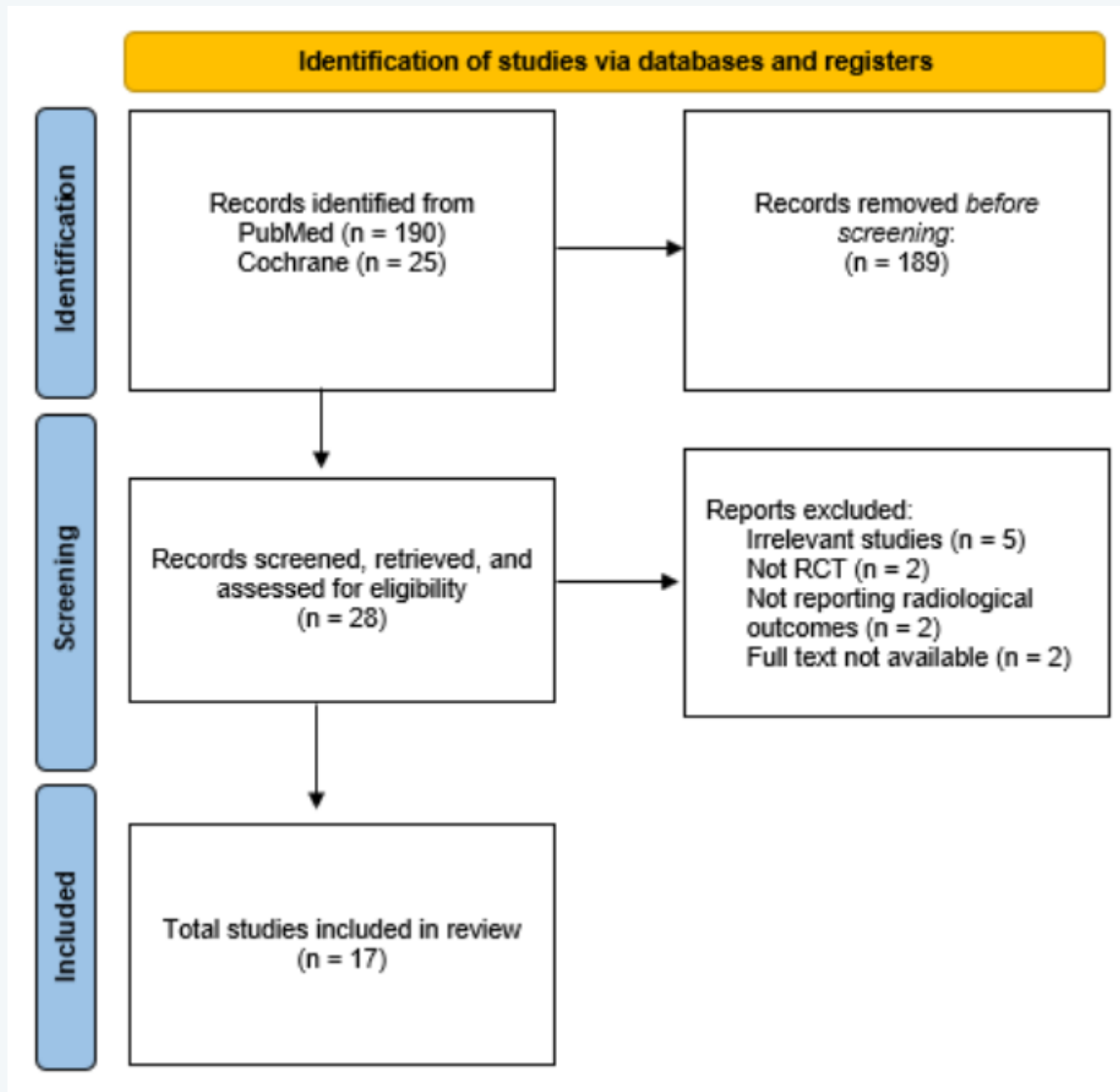
- Low molecular weight glycoproteins within TGF- β superfamily facilitate osteoinduction
- Induces the expression of 3 key osteogenic transcription factors: Runx2, Dlx5, Osx
- FDA-Approved : BMP-2 and BMP-7, commonly used recombinant human BMP (rhBMP), only for anterior lumbar interbody fusion procedures
- However, its use in other spinal fusion procedures, both lumbar and cervical, is still considered “off-label”



MATERIAL & METHODS



RESULT



- Low Risk of Bias
- Moderate Risk
- Serious Risk
- No Information

	Random sequence generation	Allocation concealment	Blinding of Participants and personnel	Blinding of outcome assessment	Selective reporting	Incomplete outcome data	Other bias
Baskin et al, 2003	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias
Burkus et al, 2003	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	Moderate Risk	No Information
Haid et al, 2004	Low Risk of Bias	Moderate Risk	Serious Risk	Low Risk of Bias	Low Risk of Bias	Moderate Risk	No Information
Vaccaro et al, 2004	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	No Information
Burkus et al, 2005	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias
Glassman et al, 2005	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Moderate Risk	Serious Risk	No Information
Dimar et al, 2006	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Moderate Risk	No Information
Glassman et al, 2008	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Moderate Risk	Low Risk of Bias	No Information
Vaccaro et al, 2008	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	No Information
Vaccaro et al, 2008	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Moderate Risk	No Information
Dawson et al, 2009	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias
Dimar et al, 2009	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk
Delawi et al, 2010	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	No Information
Hurlbert et al, 2013	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	No Information
Yan et al, 2014	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Moderate Risk
Delawi et al, 2016	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	Serious Risk	Low Risk of Bias	Moderate Risk
Cho et al, 2017	Low Risk of Bias	Low Risk of Bias	Moderate Risk	Low Risk of Bias	Low Risk of Bias	Low Risk of Bias	No Information

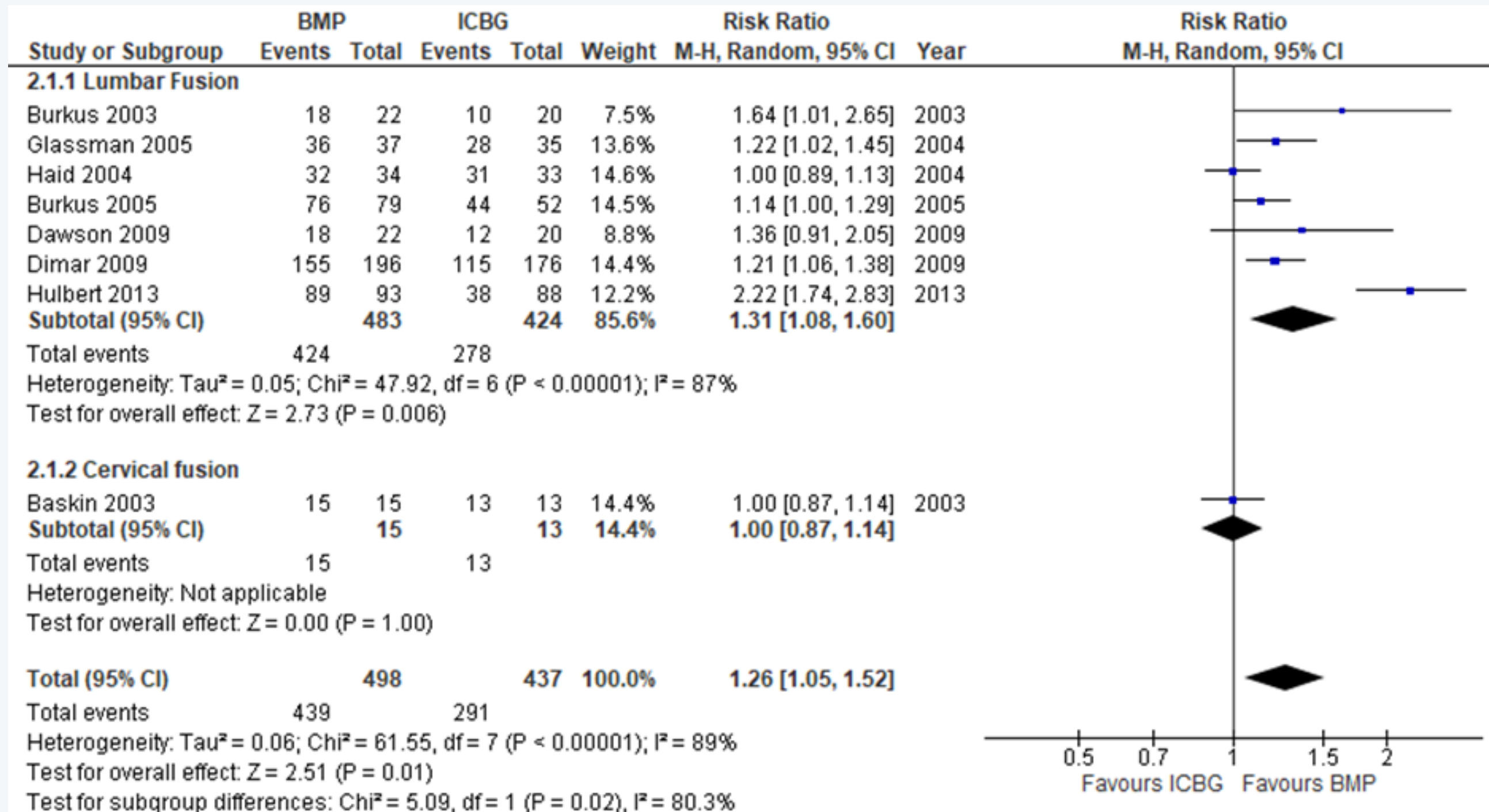
RESULT

No	Study	Study type	Preoperative diagnosis	Surgical intervention	Comparison	Sample size		Mean Age, Y		BMP Dose, mg	Follow up, mo	Outcomes
						BMP	ICBG	BMP	ICBG			
1	Baskin et al, 2003 [24]	RCT	One or 2 level cervical DDD	ACDF	rhBMP-2/ACS + Fibullar allograft vs ICBG + Fibullar allograft	18	15	51,3	47,1	NR	24	FS, neurologic status, the Neck Disability Index, SF-36, NRS, BL, LOS, OT
2	Burkus et al, 2003 [25]	RCT	Single Level Lumbar DDD, grade I spondylolisthesis	ALIF	rhBMP-2/ACS vs ICBG	22	20	41,7	44,2	4.2-8.4 mg	24	FS, reoperation
3	Haid et al, 2004 [26]	RCT	Single Level Lumbar DDD, grade I spondylolisthesis	PLIF	rhBMP-2/ACS vs ICBG	34	33	46,3	46,1	4.0-8.0 mg	24	FS, ODI, SF-36, NRS, OT, LOS, BL, complications
4	Vaccaro et al, 2004 [27]	RCT	Lumbar DDD, grade I or II spondylolisthesis and spinal stenosis	PLF	OP-1/Type I bone collagen vs ICBG	24	12	63	66	7 mg	12	FS, SF-36, ODI, LOS, OT, complications
5	Burkus et al, 2005 [28]	RCT	Single Level Lumbar DDD, Spondylolisthesis Grade 1 or less	ALIF	rhBMP-2/ACS vs ICBG	79	52	40,2	43,6	8.4-12 mg	24+	FS, ODI, SF-36, NRS, OT, LOS, BL, complications
6	Glassman et al, 2005 [29]	RCT	Lumbar DDD, grade I spondylolisthesis	PLF	rhBMP-2/CRM vs ICBG	38	36	53	53	40 mg	12	FS, OT, BL, complications
7	Dimar et al, 2006 [30]	RCT	Lumbar DDD, grade I spondylolisthesis, HNP, instability, spondylosis, stenosis	PLF	rhBMP-2/CRM vs. ICBG	53	45	50,9	52,7	40 mg	24	FS, SF-36, ODI, BL, LOS, OT, NRS, return to work, complications
8	Glassman et al, 2008 [31]	RCT	disc pathology, spondylolisthesis, stenosis, deformity, instability, post decompression revision, or adjacent level fusion	PLF	rhBMP-2/ACS vs ICBG	50	52	69,2	59,9	NR	24	FS, SF-36, ODI, NRS, BL, OT, complications

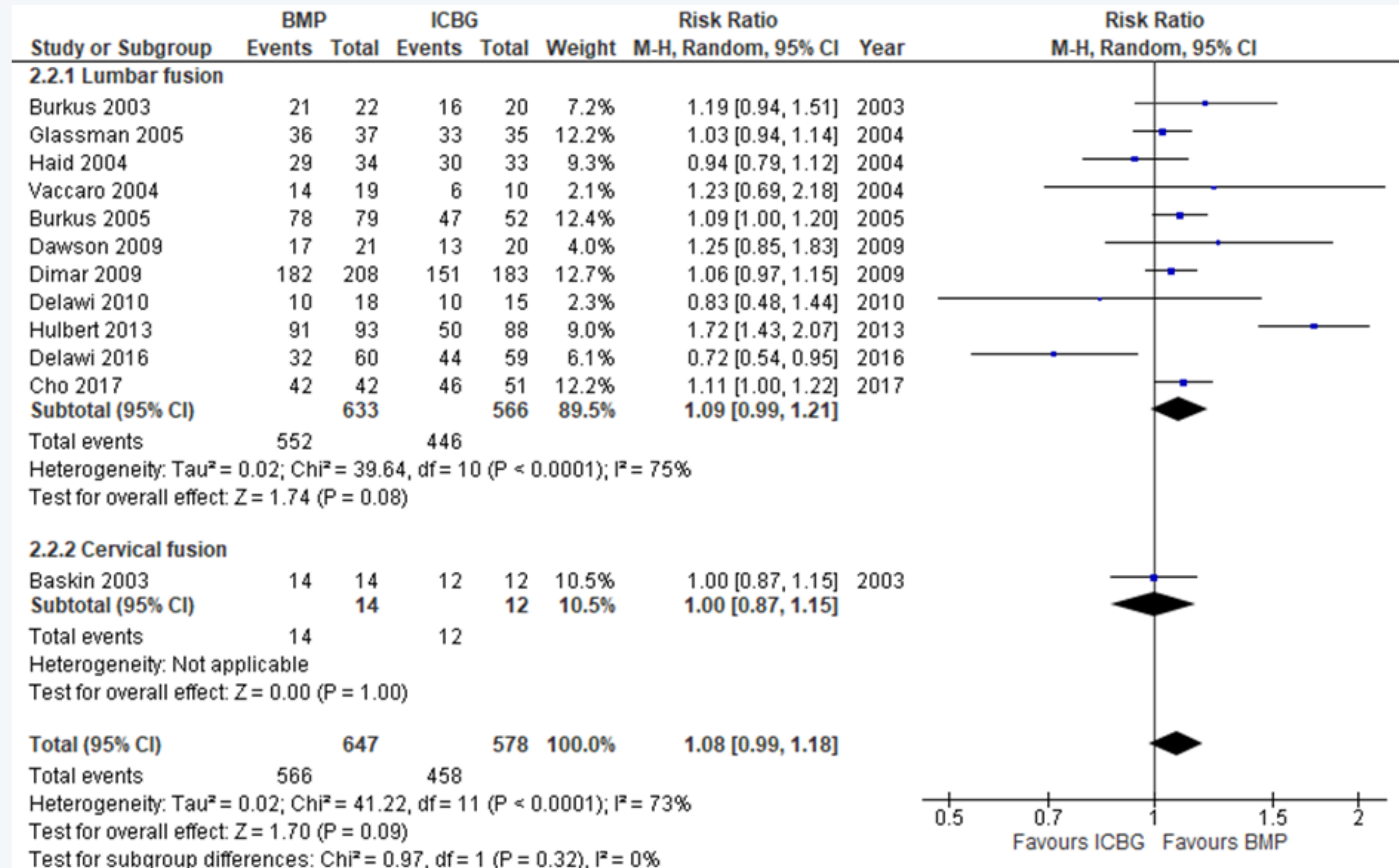
RESULT

9	Vaccaro et al, 2008 [32]	RCT	Lumbar DDD, grade I or II spondylolisthesis and spinal stenosis	PLF	OP-1/Type I bone collagen vs ICBG	208	87	68	69	7 mg	24/36+	FS, ODI, SF-36, VAS, BL, OT
10	Vaccaro et al, 2008 [33]	RCT	Lumbar DDD, grade I or II spondylolisthesis and spinal stenosis	PLF	OP-1/Type I bone collagen vs ICBG	24	12	63	67	7 mg	48	FS, ODI, complications
11	Dawson et al, 2009 [34]	RCT	Lumbar DDD, grade I spondylolisthesis	PLF	rhBMP-2/ACS vs ICBG	25	21	55,9	56,9	12 mg	24	FS, SF-36, ODI, VAS, BL, LOS, OT, complications
12	Dimar et al, 2009 [35]	RCT	Lumbar DDD, grade I spondylolisthesis, spinal instability	PLF	rhBMP-2/Bovine type I collagen vs ICBG	239	224	53,2	52,3	40 mg	24	FS, ODI, SF-36, reoperation, AD, NRS, OT, LOS, BL
13	Delawi et al, 2010 [36]	RCT	Lumbar DDD, stenosis by isthmic or degenerative spondylolisthesis	PLF	OP-1/Local autograft vs ICBG/local bone	18	16	53	55	7 mg	12	FS, ODI, AD, VAS, OT, LOS, BL
14	Hurlbert et al, 2013 [37]	RCT	One or 2 level Lumbar DDD, grade I spondylolisthesis	PLF	rhBMP-2/BCP vs ICBG	95	93	53	53	42-63 mg	48	FS, ODI, SF-36, reoperation, AD, NRS, OT, LOS, BL
15	Yan et al, 2014 [38]	RCT	Atlantoaxial instability disease	PCF	rhBMP-2/ACS + ICBG vs ICBG	68	66	68,7	69,2	NR	24	FS, VAS, JOA score, BL, LOS, OT, complications
16	Delawi et al, 2016 [39]	RCT	Lumbar DDD, grade I and II spondylolisthesis	PLF	OP-1/Type-I bone collagen + local autograft vs ICBG + local autograft	60	59	54	55	7 mg	12	FS, ODI, reoperation, AD, OT, LOS, BL
17	Cho et al, 2017 [40]	RCT	Severe Spinal Stenosis, Grade I Spondylolisthesis, Spondylosis	PLF	E. BMP-2/HA vs ICBG	42	51	64,9	62	6 mg	6	FS, SF-36, ODI, VAS, BL, LOS, OT, complications

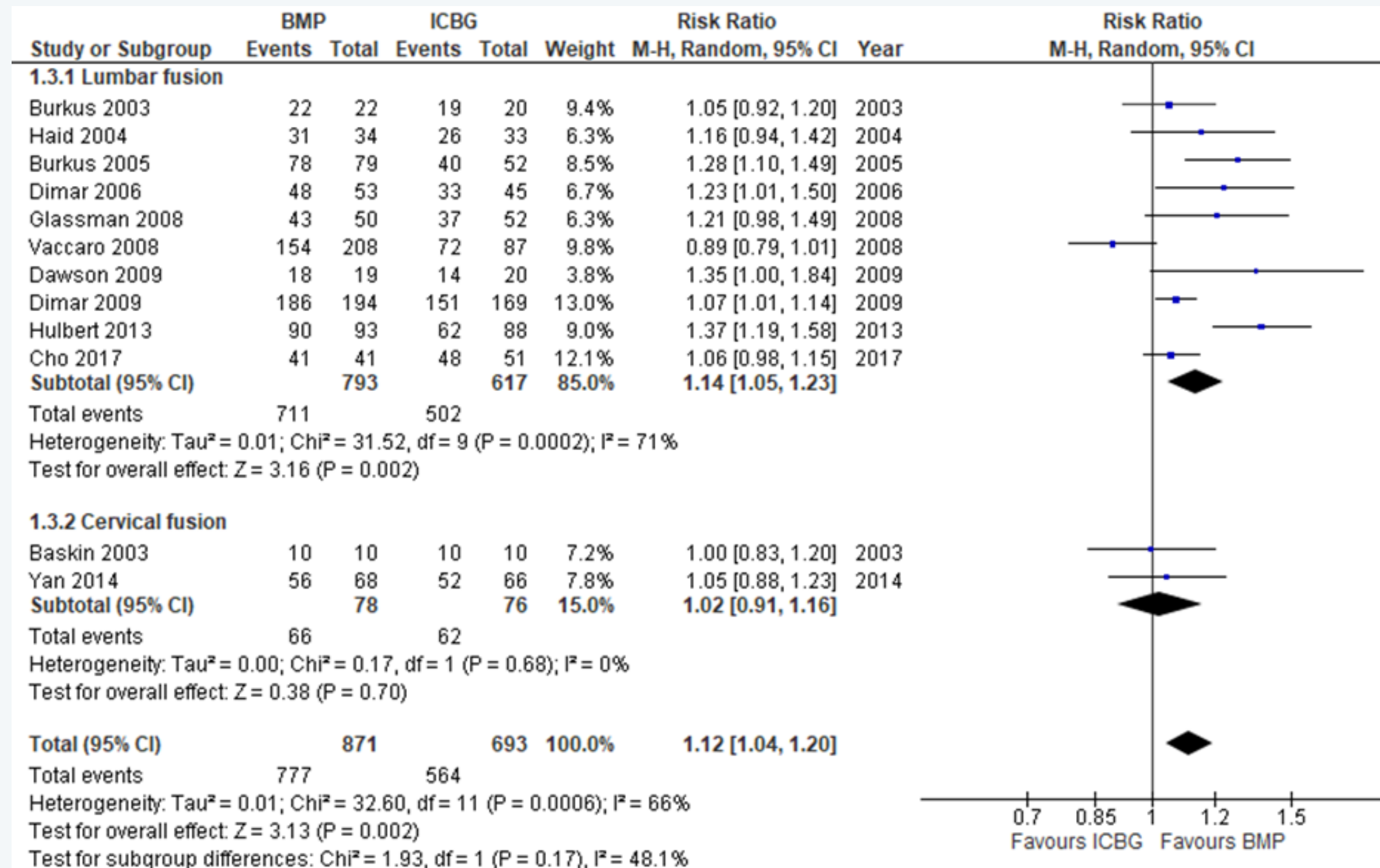
FUSION SUCCESS AT 6 MONTHS WITH SUBGROUP ANALYSIS COMPARING LUMBAR AND CERVICAL FUSION



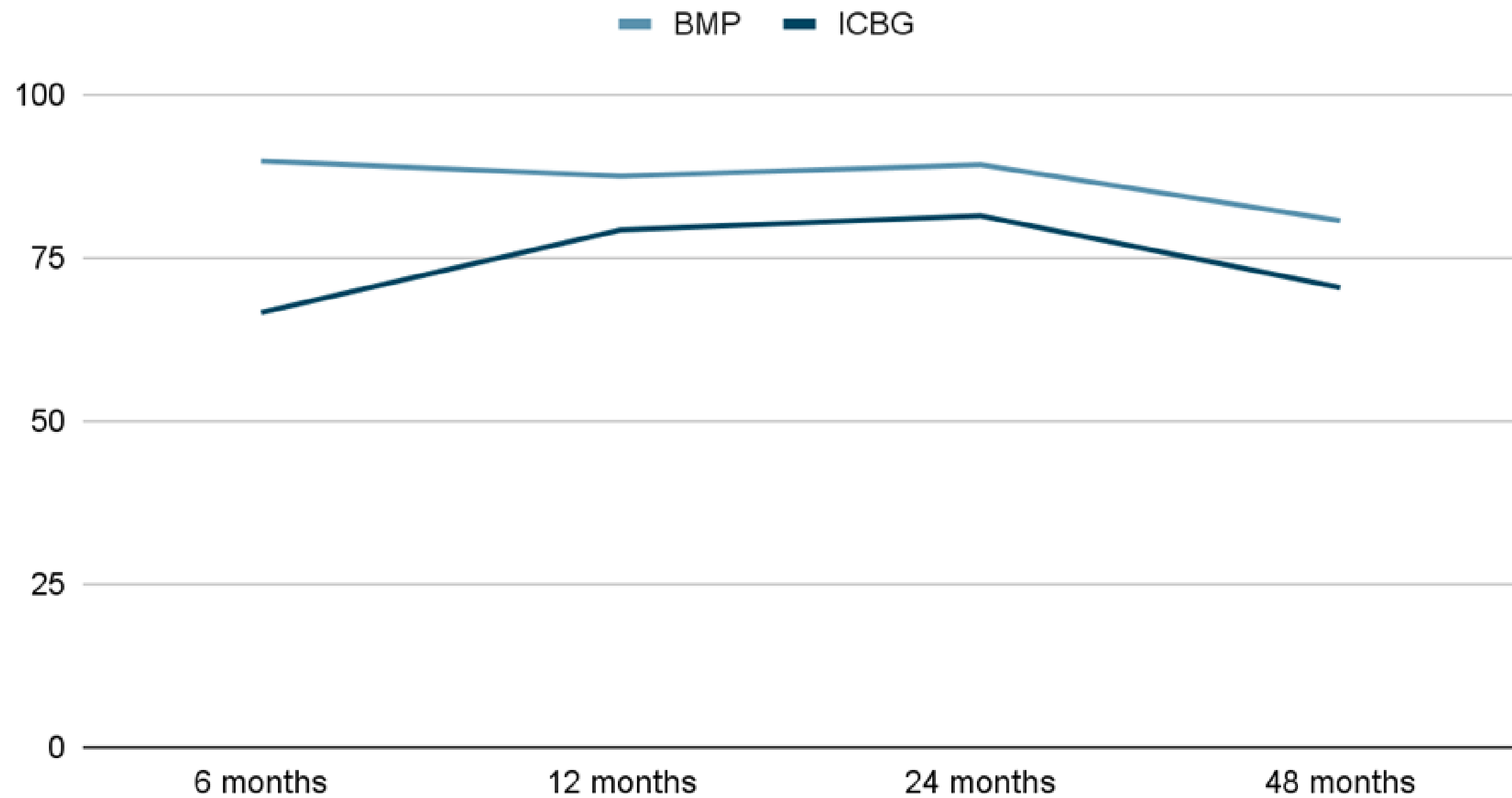
FUSION SUCCESS AT 12 MONTHS WITH SUBGROUP ANALYSIS COMPARING LUMBAR AND CERVICAL FUSION



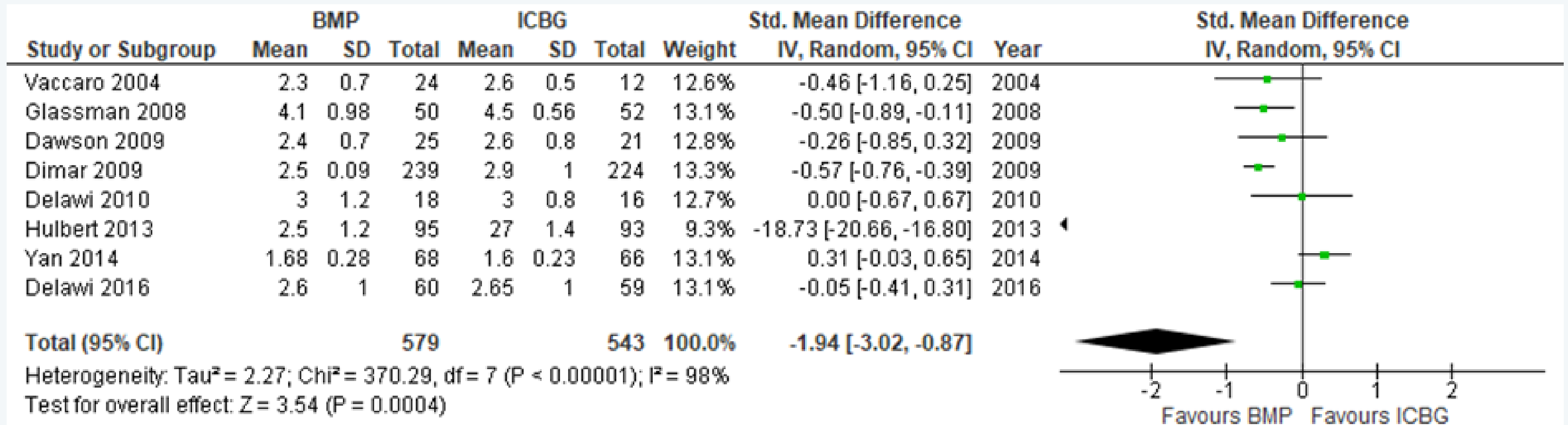
FUSION SUCCESS AT 24 MONTHS WITH SUBGROUP ANALYSIS COMPARING LUMBAR AND CERVICAL FUSION



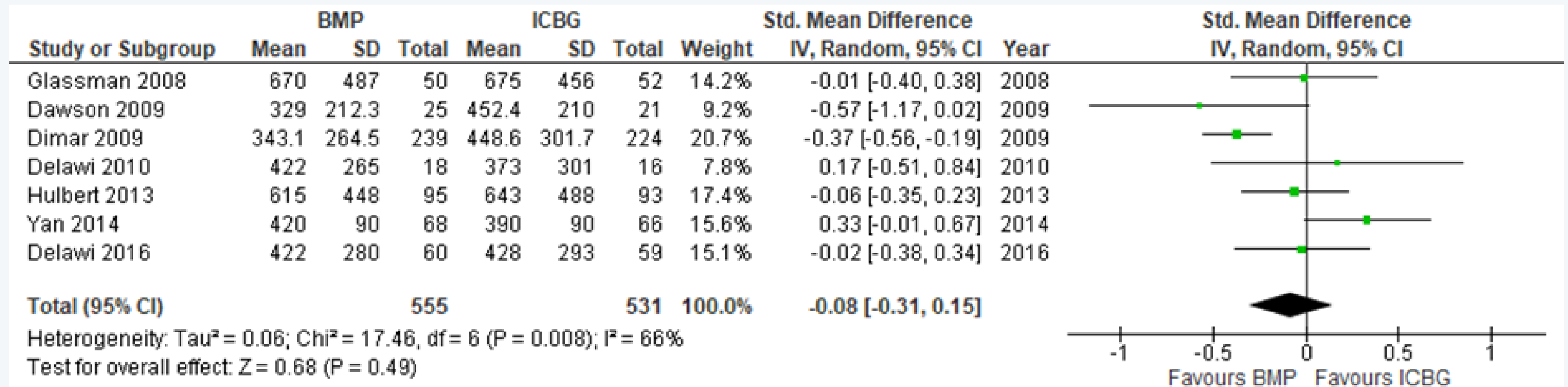
FUSION SUCCESS BETWEEN BMP (BMP-2 AND OP-1) AND ICBG OVER TIME



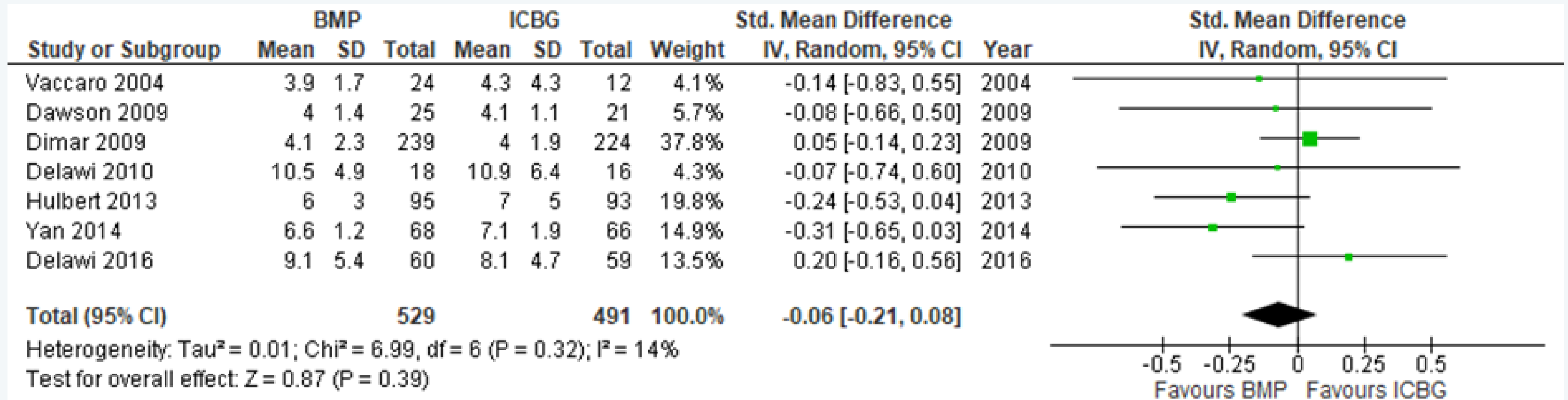
OPERATING TIME



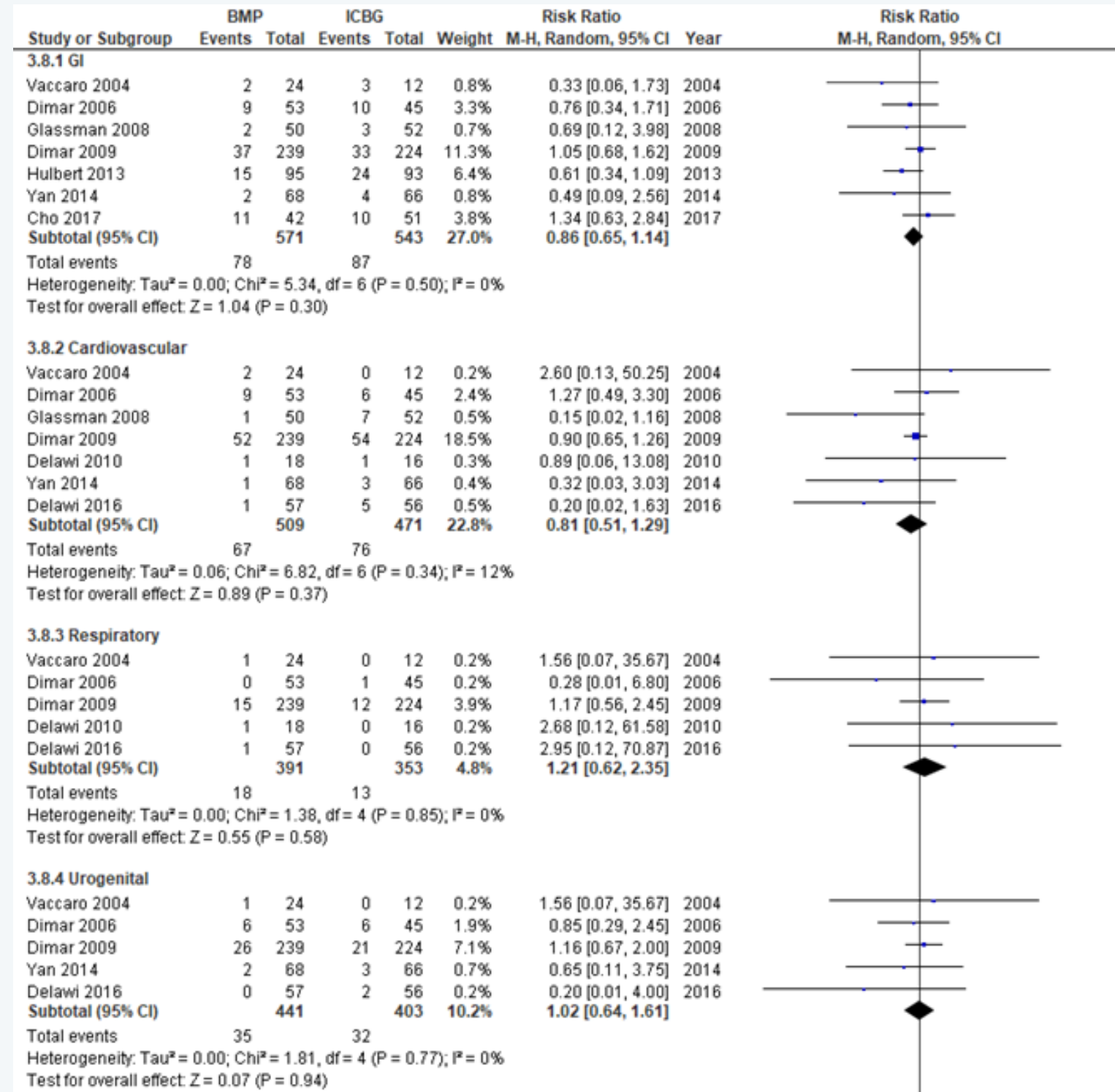
BLOOD LOSS



LENGTH OF STAY



OVERALL COMPLICATIONS



3.8.5 Non union

Burkus 2003	0	22	1	20	0.2%	0.30	[0.01, 7.07]	2003
Dimar 2006	0	53	2	45	0.2%	0.17	[0.01, 3.46]	2006
Glassman 2008	2	50	5	52	0.9%	0.42	[0.08, 2.05]	2008
Vaccaro 2008	11	38	1	6	0.6%	1.74	[0.27, 11.13]	2008
Dimar 2009	6	239	18	224	2.6%	0.31	[0.13, 0.77]	2009
Delawi 2010	2	16	2	15	0.6%	0.94	[0.15, 5.84]	2010
Hulbert 2013	2	95	2	93	0.6%	0.98	[0.14, 6.81]	2013
Delawi 2016	11	60	9	59	3.3%	1.20	[0.54, 2.69]	2016
Subtotal (95% CI)	34	573	40	514	9.1%	0.67	[0.40, 1.13]	

Total events: 34 (BMP), 40 (ICBG)
Heterogeneity: $\tau^2 = 0.04$; $\chi^2 = 7.52$, $df = 7$ ($P = 0.38$); $I^2 = 7\%$
Test for overall effect: $Z = 1.50$ ($P = 0.13$)

3.8.6 Surgical infection

Vaccaro 2004	5	24	0	12	0.3%	5.72	[0.34, 95.62]	2004
Dimar 2006	0	53	1	45	0.2%	0.28	[0.01, 6.80]	2006
Glassman 2008	1	50	4	52	0.5%	0.26	[0.03, 2.25]	2008
Dawson 2009	1	25	2	21	0.4%	0.42	[0.04, 4.31]	2009
Delawi 2010	1	18	1	16	0.3%	0.89	[0.06, 13.08]	2010
Hulbert 2013	11	95	5	93	2.1%	2.15	[0.78, 5.96]	2013
Delawi 2016	5	57	5	56	1.5%	0.98	[0.30, 3.21]	2016
Subtotal (95% CI)	24	322	18	295	5.3%	1.15	[0.60, 2.20]	

Total events: 24 (BMP), 18 (ICBG)
Heterogeneity: $\tau^2 = 0.01$; $\chi^2 = 6.10$, $df = 6$ ($P = 0.41$); $I^2 = 2\%$
Test for overall effect: $Z = 0.41$ ($P = 0.68$)

3.8.7 Req revision

Burkus 2003	0	22	1	20	0.2%	0.30	[0.01, 7.07]	2003
Glassman 2005	1	38	3	36	0.4%	0.32	[0.03, 2.90]	2004
Haid 2004	6	34	6	33	2.0%	0.97	[0.35, 2.71]	2004
Burkus 2005	2	79	8	52	0.9%	0.16	[0.04, 0.74]	2005
Dimar 2006	0	53	3	45	0.3%	0.12	[0.01, 2.29]	2006
Vaccaro 2008	21	162	11	66	4.8%	0.78	[0.40, 1.52]	2008
Dimar 2009	20	239	36	224	8.0%	0.52	[0.31, 0.87]	2009
Dawson 2009	2	22	2	18	0.6%	0.82	[0.13, 5.25]	2009
Delawi 2010	0	18	1	15	0.2%	0.28	[0.01, 6.43]	2010
Delawi 2016	10	60	2	59	1.0%	4.92	[1.12, 21.49]	2016
Cho 2017	0	41	1	51	0.2%	0.41	[0.02, 9.87]	2017
Subtotal (95% CI)	62	768	74	619	18.7%	0.64	[0.39, 1.06]	

Total events: 62 (BMP), 74 (ICBG)
Heterogeneity: $\tau^2 = 0.17$; $\chi^2 = 14.26$, $df = 10$ ($P = 0.16$); $I^2 = 30\%$
Test for overall effect: $Z = 1.75$ ($P = 0.08$)

3.8.8 Tumor/Malignancy

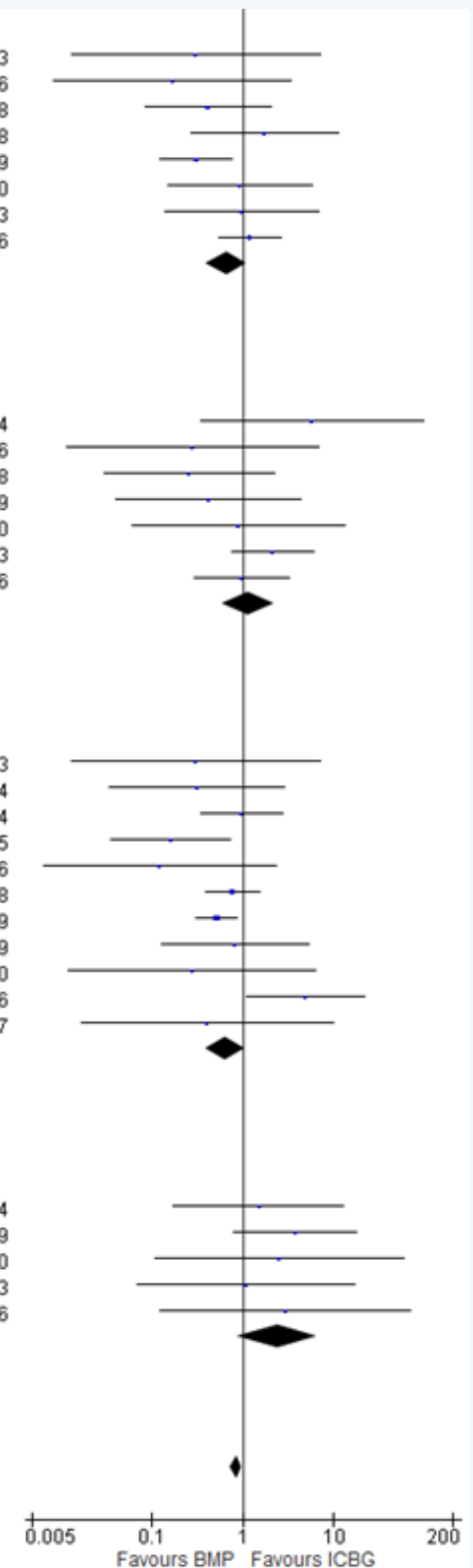
Vaccaro 2004	3	24	1	12	0.5%	1.50	[0.17, 12.94]	2004
Dimar 2009	8	239	2	224	0.9%	3.75	[0.80, 17.47]	2009
Delawi 2010	1	18	0	15	0.2%	2.53	[0.11, 57.83]	2010
Hulbert 2013	1	85	1	93	0.3%	1.09	[0.07, 17.22]	2013
Delawi 2016	1	57	0	56	0.2%	2.95	[0.12, 70.87]	2016
Subtotal (95% CI)	14	423	4	400	2.1%	2.42	[0.88, 6.69]	

Total events: 14 (BMP), 4 (ICBG)
Heterogeneity: $\tau^2 = 0.00$; $\chi^2 = 0.84$, $df = 4$ ($P = 0.93$); $I^2 = 0\%$
Test for overall effect: $Z = 1.71$ ($P = 0.09$)

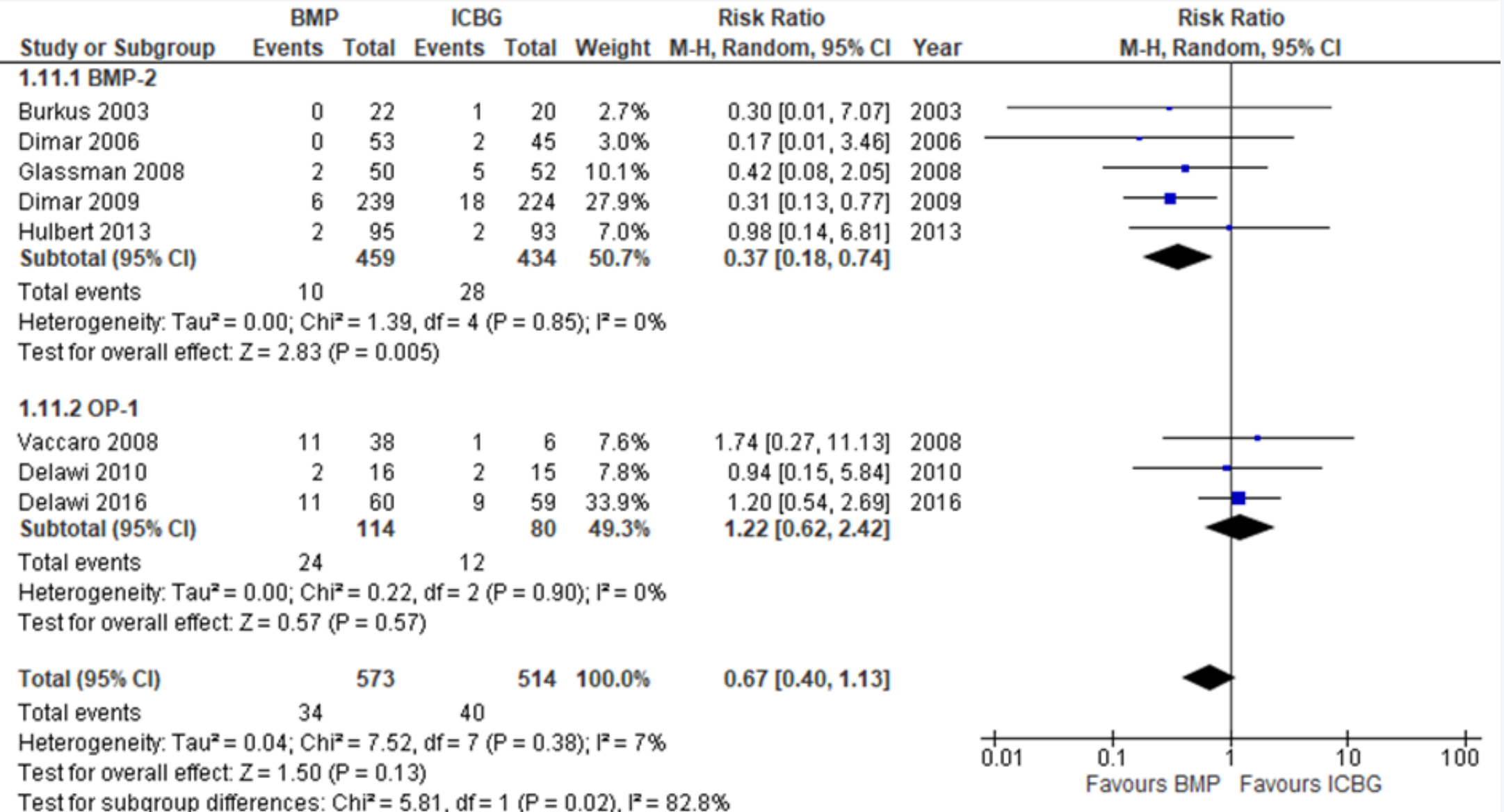
Total (95% CI) 3998 3598 100.0% 0.85 [0.74, 0.99]

Total events: 332 (BMP), 344 (ICBG)
Heterogeneity: $\tau^2 = 0.00$; $\chi^2 = 54.23$, $df = 54$ ($P = 0.47$); $I^2 = 0\%$
Test for overall effect: $Z = 2.11$ ($P = 0.04$)

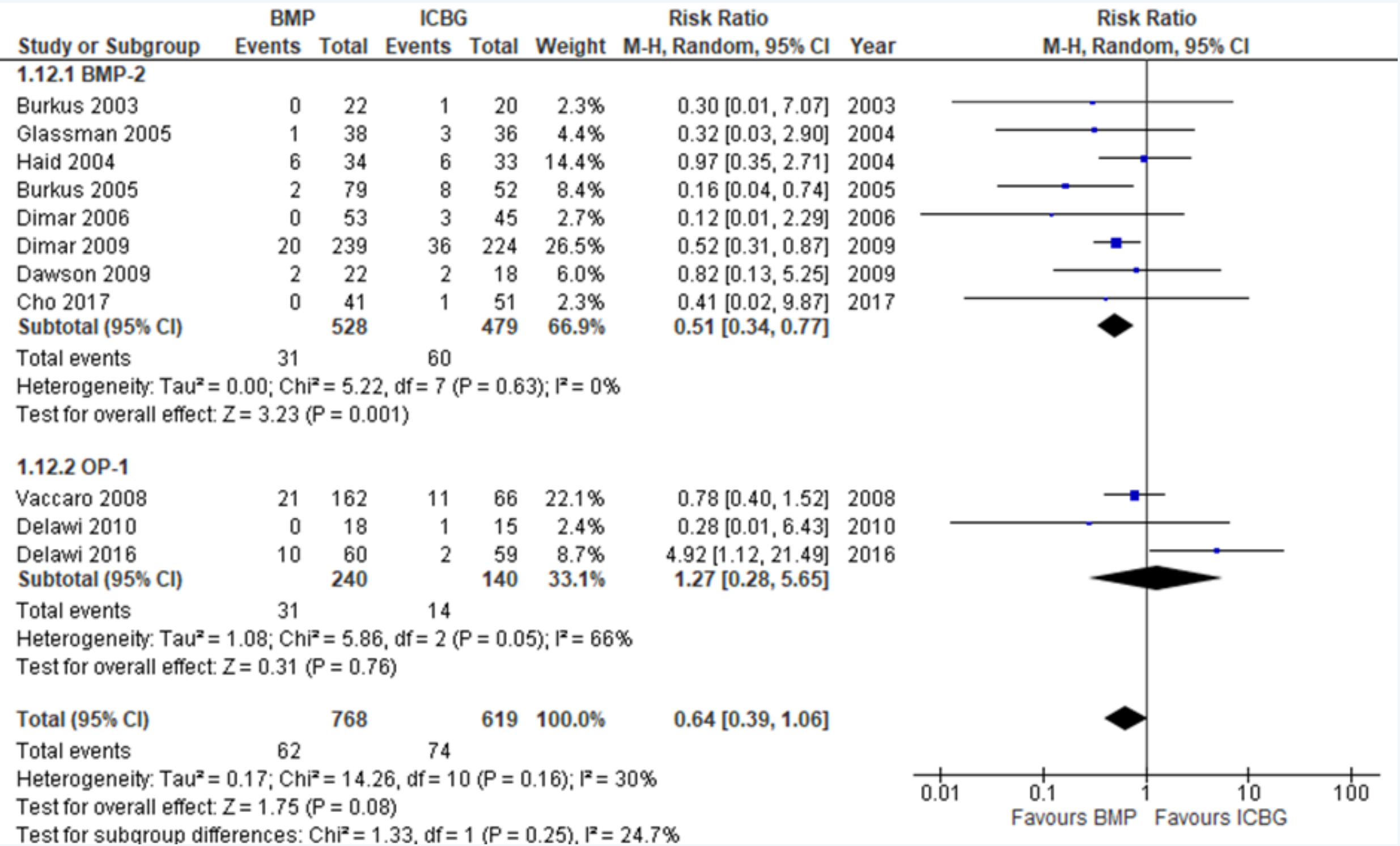
Test for subgroup differences: $\chi^2 = 8.43$, $df = 7$ ($P = 0.30$), $I^2 = 17.0\%$



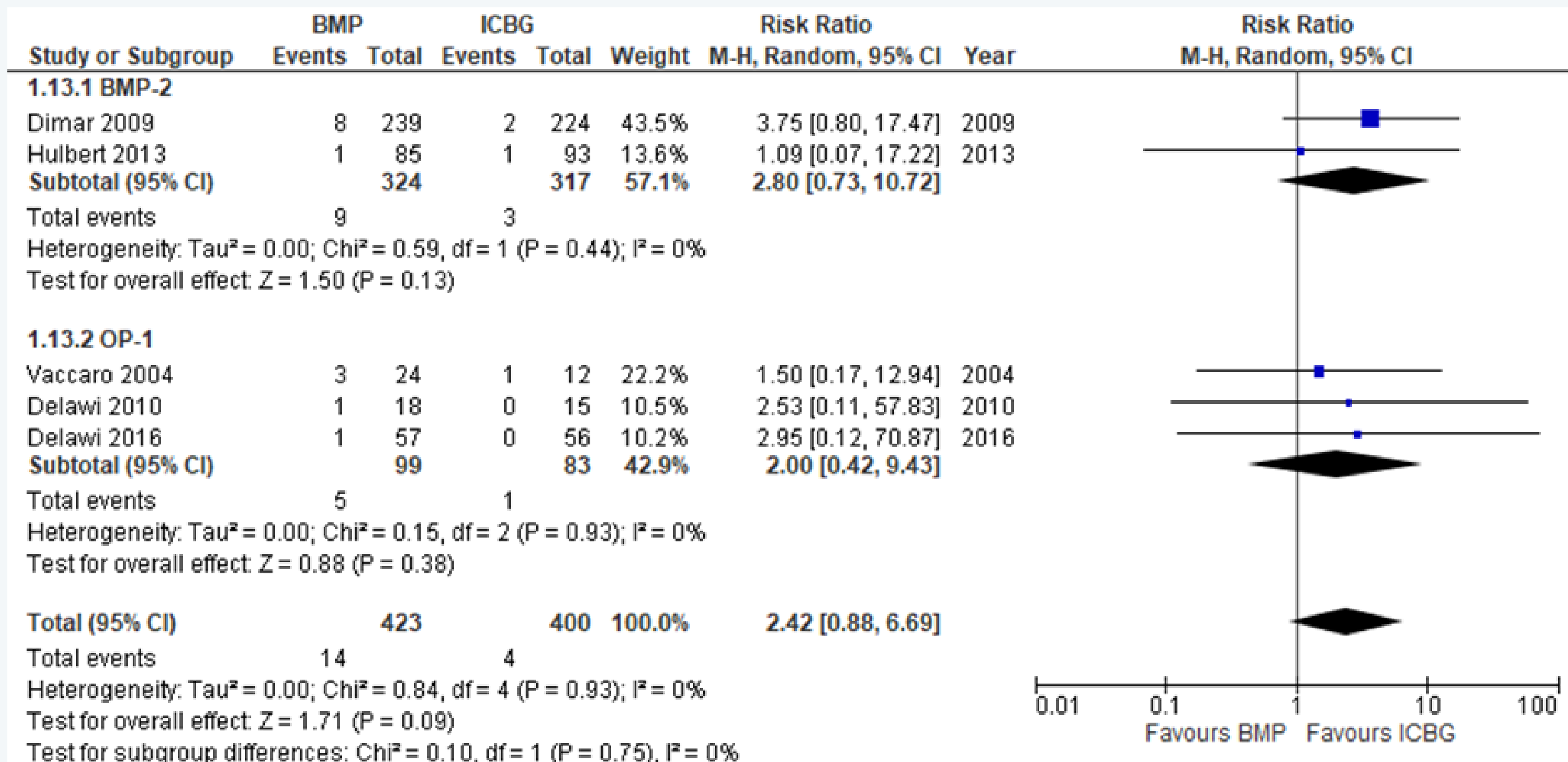
CASES OF NON-UNION IN BMP-2 VS OP-1



CASES REQUIRING REVISION SURGERY IN BMP-2 VS OP-1



MALIGNANCY IN BMP-2 VS OP-1



DISCUSSION

- BMP showed higher and faster fusion rates, shorter operative times, better functional outcomes, and fewer complications compared to ICBG.
- Fusion Success Rates:
- BMP-2 had significantly higher fusion success rates, especially in lumbar fusion.
- Cervical fusion did not show significant differences between BMP and ICBG.
- Clinical Outcomes: BMP group had better scores in SF-36, ODI, back pain, and leg pain, indicating superior clinical outcomes.
- Operative Time: BMP group had significantly shorter operative times, reducing the risk of surgical site infection.
- Complication Rates: BMP exhibited lower complication rates and fewer nonunions and revision surgeries compared to ICBG.

DISCUSSION

- BMP's advantages were mostly observed with BMP-2, particularly in lumbar fusion.
- BMP use in cervical fusion showed mixed results, with some studies suggesting an increased risk of complications in anterior cervical fusion.
- There is an ongoing debate about BMP's potential link to malignancy, though evidence is not conclusive.

Study Limitations

Limited cervical fusion data and high heterogeneity in study results.

Influencing factors like age, BMP dosage, and carrier type were not fully considered.

CONCLUSION

BMP-2 in lumbar fusion shows superior outcomes to ICBG, with better fusion success rates evident as early as 6 months, along with fewer non-union complications and revision surgeries. The efficacy and safety of using BMP in cervical fusion and OP-1 remain inconclusive. Further investigation, particularly with OP-1, is warranted for BMP's application in cervical fusion, necessitating studies with larger sample sizes.

Future Research:

Larger studies, especially on cervical fusion, and research on BMP dosage, carrier type, and cost-effectiveness are needed to clarify BMP's role in spinal fusion.

THANK YOU

